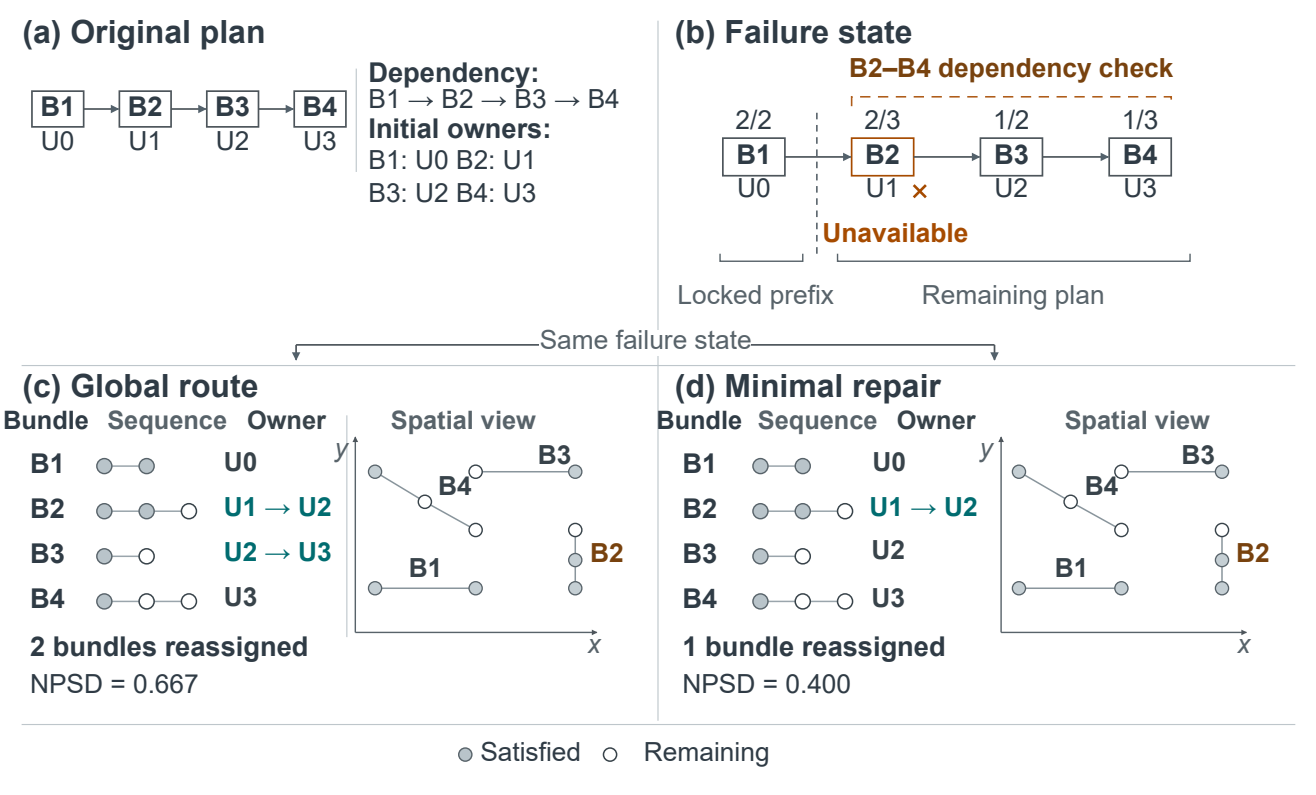

(a) Original plan
B1
U0
B2
U1
B3
U2
B4
U3
Dependency:
B1 → B2 → B3 → B4
Initial owners:
B1: U0 B2: U1
B3: U2 B4: U3
(b) Failure state
B2–B4 dependency check
2/2
2/3
1/2
1/3
B1
U0
B2
U1 ×
B3
U2
B4
U3
Unavailable
Locked prefix
Remaining plan
Same failure state
(c) Global route
Bundle Sequence Owner
B1 U0
B2 U1 → U2
B3 U2 → U3
B4 U3
2 bundles reassigned
NPSD = 0.667
Spatial view
y
x
B1
B2
B3
B4
(d) Minimal repair
Bundle Sequence Owner
B1 U0
B2 U1 → U2
B3 U2
B4 U3
1 bundle reassigned
NPSD = 0.400
Spatial view
y
x
B1
B2
B3
B4
Satisfied
Remaining